

# Spectra Sleuth Showdown

Can two white lights hide different color fingerprints?

**Win condition:** Most correct spectrum matches and strongest safe fireworks-science explanation.

## THE SCIENCE YOU ARE USING

**Diffraction splits light.** A diffraction grating bends different colors by different amounts, spreading light into a spectrum. Two lights that look the same white can have very different spectra: a smooth rainbow from a hot filament, a broad colored band from a white LED, or separated bright lines from a neon or gas source.

**Spectra as fingerprints.** Each light source has a characteristic spectrum, its fingerprint. This is the same science behind firework colors, where different metal salts emit specific colors. You match mystery sources to clue cards using their spectra, no flame needed.

## HOW TO BUILD AND RUN IT

- 01 Look through the grating.** View a white light through the grating and watch it spread into colors.
- 02 Compare two sources.** Look at a white LED and a neon light. The LED shows a broad band; the neon shows separate bright lines.
- 03 Sketch the spectra.** Draw what you see for each mystery source: continuous band or separate lines.
- 04 Match to clue cards.** Use your sketches to match each source to its spectrum card.
- 05 Connect to fireworks.** Explain how this links to the colors in a firework display, safely.

**Safety:** No flame tests. No lasers. Diffraction glasses are not eye protection. Allergy: None.

## MATERIALS

|                                   |              |
|-----------------------------------|--------------|
| Diffraction glasses or gratings   | per student  |
| White LED and neon or gas sources | shared       |
| Incandescent source               | if available |
| Spectrum cards                    | set          |
| Museum clue sheets                | per team     |

## YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): \_\_\_\_\_

The ONE variable we are testing: \_\_\_\_\_

Baseline result: \_\_\_\_\_

After redesign: \_\_\_\_\_

## DEFEND YOUR DESIGN

What evidence made you change your design? \_\_\_\_\_

If you ran it again, what is the ONE thing you would change? \_\_\_\_\_

## HOW YOU SCORE (100 POINTS)

| SCORING CRITERION  | POINTS |
|--------------------|--------|
| Correct matches    | 40     |
| Spectrum sketches  | 20     |
| Explanation        | 20     |
| Exhibit connection | 20     |
| Total              | 100    |